

KYSHTOVKA, Russian Federation

WMO: 294050

Lat: 56.6011N

Lon: 76.5613E

Elev: 381

StdP: 14.49

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.7	-29.2	-40.3	0.5	-34.4	-35.0	0.8	-29.0	20.3	11.4	17.7	13.5	3.3	0	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	84.1	66.6	81.3	65.4	78.3	64.0	69.9	79.7	67.8	77.6	66.0	75.0	7.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.1	74.5	64.2	91.2	72.7	62.5	85.7	70.7	34.1	79.7	32.4	77.8	31.0	75.1	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
20.9	17.6	15.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-40.8	88.9	7.3	3.1	-46.1	91.1	-50.4	92.9	-54.5	94.6	-59.8	96.8		
			-41.1	73.3	6.9	3.4	-46.1	75.8	-50.1	77.7	-54.0	79.7	-59.0	82.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	-3.0	3.5	19.2	36.7	50.8	61.6	64.9	59.5	49.0	35.9	17.1	3.7	(6)
(7)		DBStd	25.90	15.03	13.50	10.85	10.44	9.87	7.80	6.11	6.68	7.76	9.08	14.60	15.33	(7)
(8)		HDD50	7417	1642	1301	954	413	115	9	0	5	111	443	988	1436	(8)
(9)		HDD65	11692	2107	1721	1419	848	449	151	78	196	483	901	1438	1901	(9)
(10)		CDD50	1358	0	0	0	15	140	357	463	298	79	6	0	0	(10)
(11)		CDD65	161	0	0	0	0	9	49	76	25	2	0	0	0	(11)
(12)		CDH74	1780	0	0	0	2	200	571	691	283	33	0	0	0	(12)
(13)		CDH80	378	0	0	0	0	43	125	153	52	5	0	0	0	(13)
(14)	Wind	WSAvg	6.9	6.1	6.6	7.7	8.1	8.4	6.4	6.1	6.6	7.2	7.3	6.8	(14)	
(15)	Precipitation	PrecAvg	16.9	1.0	0.7	0.7	0.8	1.5	2.1	2.3	2.3	1.4	1.5	1.4	1.2	(15)
(16)		PrecMax	23.5	2.0	1.4	2.0	1.7	3.6	5.4	4.2	4.8	3.1	2.8	2.6	2.5	(16)
(17)		PrecMin	9.6	0.0	0.0	0.0	0.0	0.4	0.6	0.2	0.4	0.3	0.6	0.4	0.4	(17)
(18)		PrecStd	3.1	0.6	0.4	0.6	0.6	0.8	1.1	1.0	1.1	0.8	0.7	0.5	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	34.0	43.7	71.0	85.0	87.1	87.8	85.1	78.2	64.5	43.9	34.2	(19)
(20)			MCWB	28.8	31.2	36.8	53.5	63.0	66.2	71.3	67.5	60.3	50.7	40.2	33.0	(20)
(21)		2%	DB	27.0	30.7	39.9	63.2	79.5	83.7	83.9	80.8	72.3	58.8	39.8	31.6	(21)
(22)			MCWB	25.7	28.8	34.4	50.1	59.2	66.1	67.8	65.5	58.1	48.1	37.6	30.3	(22)
(23)		5%	DB	23.4	27.3	36.9	57.8	75.0	80.5	81.1	77.1	67.0	53.3	36.1	28.0	(23)
(24)			MCWB	22.2	25.8	33.1	47.1	57.3	65.4	66.5	64.2	55.8	45.6	34.4	26.8	(24)
(25)		10%	DB	19.4	23.2	34.1	52.4	69.6	77.1	77.9	73.0	62.4	48.9	33.3	24.1	(25)
(26)			MCWB	18.4	21.6	31.1	43.8	54.8	63.8	65.5	62.3	53.3	43.7	31.9	23.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.3	31.5	37.7	55.4	65.4	72.8	73.3	69.5	62.5	52.2	40.5	33.4	(27)
(28)			MCDWB	30.4	33.1	42.4	65.8	80.2	81.2	82.7	79.9	74.3	61.4	42.4	33.9	(28)
(29)		2%	WB	25.6	28.8	35.2	51.2	61.1	69.4	70.4	67.3	59.2	49.1	38.0	30.4	(29)
(30)			MCDWB	26.9	30.6	38.6	61.8	76.2	79.0	80.5	77.7	68.8	56.7	39.8	31.7	(30)
(31)		5%	WB	22.2	25.8	33.4	47.9	58.4	67.0	68.4	65.4	56.6	46.4	34.5	26.9	(31)
(32)			MCDWB	23.2	27.3	36.5	56.8	72.0	77.0	78.0	74.3	65.7	52.1	35.6	28.0	(32)
(33)		10%	WB	18.4	21.7	31.2	44.5	55.7	64.8	66.6	63.2	54.1	43.8	32.3	23.0	(33)
(34)			MCDWB	19.4	23.1	34.2	51.9	67.9	74.5	75.1	71.5	61.4	48.7	33.3	24.1	(34)
(35)	Mean Daily Temperature Range	MDBR	15.2	17.2	17.6	17.5	21.9	20.6	18.9	19.2	17.7	13.3	11.9	14.2	(35)	
(36)		5% DB	MCDBR	14.1	13.3	16.8	25.0	30.1	26.7	24.1	26.0	26.5	21.9	10.5	13.4	(36)
(37)			MCWBR	13.8	12.4	12.8	14.9	14.0	12.8	11.3	12.6	13.7	13.0	9.0	13.0	(37)
(38)		5% WB	MCDBR	14.1	13.0	14.7	23.4	27.7	23.1	21.5	22.9	24.0	19.8	10.0	13.2	(38)
(39)			MCWBR	13.8	12.4	11.8	14.5	15.0	12.4	11.1	12.2	13.3	12.6	8.8	13.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.311	0.368	0.408	0.400	0.406	0.428	0.414	0.364	0.349	0.309	0.280	(40)	
(41)		taud	2.029	2.015	1.928	2.062	2.238	2.297	2.231	2.247	2.346	2.230	2.074	1.971	(41)	
(42)		Ebn at Noon	182	226	240	253	265	265	255	250	248	213	166	140	(42)	
(43)		Edh at Noon	20	31	44	45	40	38	40	37	29	25	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	250	579	1063	1437	1638	1834	1677	1305	895	438	204	153	(44)	
(45)		RadStd	23	38	66	124	152	180	104	140	90	42	36	22	45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page